Luis A. Rodriguez Condit

Technical Architect – Embedded & Web Systems

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Value Proposition

- Software architect with 20+ years building systems that bridge embedded hardware and modern web applications
- Lead and mentor teams end-to-end: architect solutions, write code, coordinate delivery, develop engineers
- Proven innovator: WebAssembly adoption, 70% build time reduction, accessibility implementation
- **Technical Profile**
    - **Expert (10+ years):** JavaScript • TypeScript • C • Java • Node.js • HTML5 • CSS • Git
    - **Proficient: (~5 years)** Angular • REST APIs • CI/CD • Emscripten/WebAssembly
    - **Currently Expanding: (<2 years)** React • Spring Boot • Docker containerization • AWS
- Built calculator emulators and web applications used by millions of students globally
- Elected to Texas Instruments' Technical Ladder for technical excellence

Skills

- **Systems & Emulation:** Hardware emulation (Z80, eZ80, T4, Lapis MCU), WebAssembly, embedded systems, communication protocols
- **Frontend & UX:** Responsive web design, cross-browser development, WebUSB/WebSerial, accessibility (WCAG)
- **DevOps & Testing:** Jenkins, Gulp, CMake, unit testing (Mocha/Chai/Jasmine/Jest), e2e testing (Selenium), build automation
- **Architecture:** RESTful services, SPA architecture, API development, object-oriented design
- **Leadership:** Technical leadership, mentorship, Agile/Scrum, code review, cross-functional collaboration

Professional Experience

Texas Instruments -- Software Architect -- *(Nov 2010 – Sep 2025, Dallas, TX)*

Built desktop and web applications bridging low-level hardware emulation and proprietary communication protocols, bringing embedded calculator systems to millions of students and educators. Led technical planning and project execution for multiple development teams. Recognized with election to TI's Technical Ladder for technical excellence.

Web Applications

- Architected and led development of multiple SPAs using TypeScript and Angular, serving over 3.5 million users globally.
- Designed and implemented license activation system with Angular frontend, RESTful middleware (Java/Spring Boot), and Oracle database backend.
- Defined front-end architecture and component-based design patterns across calculator application iterations, ensuring responsive web design and cross-platform user experiences.
- Implemented OAuth single-sign-on authentication and third-party license validation for online calculator applications.
- Owned full development lifecycle, translated business requirements into technical specifications and Jira epics, led Scrum execution, coordinated across QA and product teams.

Hardware Emulation & Integration

- Built high-performance calculator emulators for the browser, implementing Z80, eZ80, Toshiba T4, and Lapis MCU architectures in pure JavaScript for maximum browser compatibility.
- Led WebAssembly adoption for major calculator platform, cross-compiling C code to run natively in the browser.
- Integrated WebUSB and WebSerial APIs into web applications, enabling direct browser-to-calculator communication without native drivers or plugins.
- Designed and implemented JavaScript/C glue layers for cross-compiling embedded libraries with Emscripten.

Accessibility & User Experience

- Implemented screen reader support across Z80 assembly and JavaScript layers, enabling vision-impaired users to interact with calculator emulators.
- Partnered with product management and UX designers to translate complex requirements into user-centric interfaces.

Technical Leadership & DevOps

- Reduced build times by 70% through Jenkins parallelization through gulp workflow optimization.
- Built automated testing infrastructure including WebSocket-driven harness for Robot Framework and Python test reuse. Prototyped automation framework with Node.js, Cucumber, Mocha and Selenium.
- Raised code quality standards across multiple teams through mandatory code reviews, unit testing (Mocha/Chai/Jasmine), and static analysis practices.
- Accelerated technology adoption by delivering rapid prototypes that demonstrated feasibility and de-risked technical decisions.
- Mentored junior and mid-level engineers through code reviews, technical deep-dives, and hands-on guidance, fostering team growth and morale.

Digital Chocolate -- Game Engineering Manager -- *(Jun 2009 – Oct 2010, Mexicali, Mexico)*

- Managed teams responsible for porting mobile games across various US carrier devices, focusing on technical quality and resource allocation. Mentored the engineering team through hands-on tutorials and code deep-dives, directly contributing to team skill development.
- Created a client module for Qualcomm's Application Value Billing API (C/C++) to enable in-app purchases and promote code reuse across future game titles.

LemonQuest -- iPhone Programmer -- *(Apr 2008 – Jun 2009, Salamanca, Spain)*

- Ported games from DirectX to iPhone's OpenGL ES, optimizing the user experience by translating desktop inputs for touch and accelerometer use.

Gameloft -- Game Programmer -- *(Aug 2005 – Mar 2008, Mexicali, Mexico)*

- Led the team that developed the first successful iPhone game prototype for the studio, demonstrating rapid feature development and adaptability to emerging mobile/browser platforms.
- Led the technical effort to port multiple game titles to resource-limited, 3D-capable mobile devices using Java/J2ME and OpenGL ES, focusing on maintaining graphic fidelity and stability within strict memory constraints.
- Directed and participated in team efforts for porting games to severely space-constrained devices (Virgin Mobile catalog). This required strategic scope reduction, asset compression, and intensive micro-optimizations to ensure acceptable performance and quality within limited runtime memory.

Education

Master's Coursework in Electronics and Telecommunications

Centro de Investigación Científica y de Educación Superior de Ensenada (CICESE)

Bachelor of Science in Electronics Systems Engineering

Instituto Tecnológico y de Estudios Superiores de Monterrey (ITESM)